



KUWAIT UNIVERSITY
Faculty of Science
Department of Mathematics

Math 101: Calculus I

First Exam, 25 October 2018

Duration: 90 minutes

- *In questions 1-8, show all your work.*
- *Calculators and communication devices are not allowed in the examination room.*

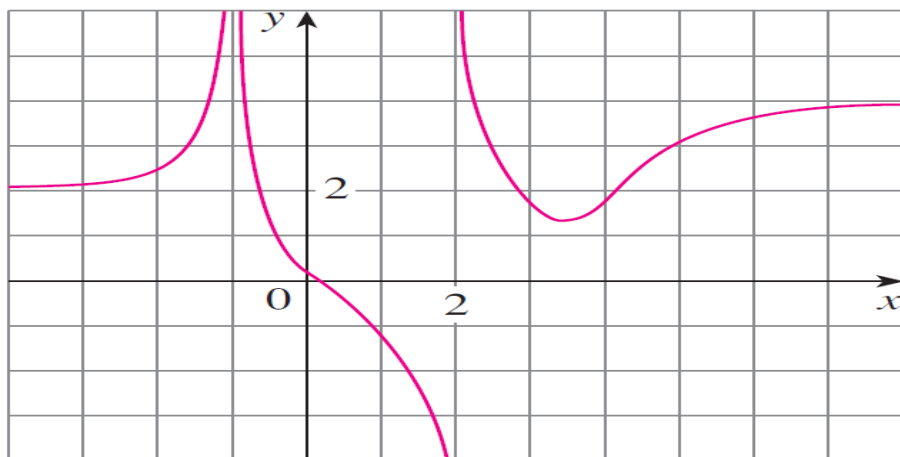
Name: _____

Student ID: _____

Section Number: _____ Serial Number: _____

Q1 (10 pts)	
Q2 (10 pts)	
Q3 (15 pts)	
Q4 (20 pts)	
Q5 (10 pts)	
Q6 (15 pts)	
Q7 (10 pts)	
Q8 (10 pts)	
Total	

1. [10 pts.] Use the given graph of the function f shown below to find all vertical and horizontal asymptotes to the graph of f .



2. [10 pts.] Find the exact value of $\sin \left(\sec^{-1} \left(\frac{5}{3} \right) \right)$.

3. [10 + 5 = 15 pts.] Let $f(x) = \ln\left(\frac{2}{x} + 3\right)$.

(a) Find the inverse function of f .

(b) Find the range of f .

4. [10 + 10 = 20 pts.] Evaluate each of the following limits, if it exists.

(a) $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x^2 - 9x}$

(b) $\lim_{x \rightarrow 1} |x - 1| \sin\left(\frac{1}{x^2 - 1}\right)$

5. [10 pts.] Find the value of the number A , if any, that makes the function f continuous at $x = 0$.

$$f(x) = \begin{cases} \frac{\frac{1}{(x+2)^2} - \frac{1}{4}}{x}, & \text{if } x \neq 0, \\ 2 - A \sec x, & \text{if } x = 0. \end{cases}$$

6. [10 + 5 = 15 pts.] Let

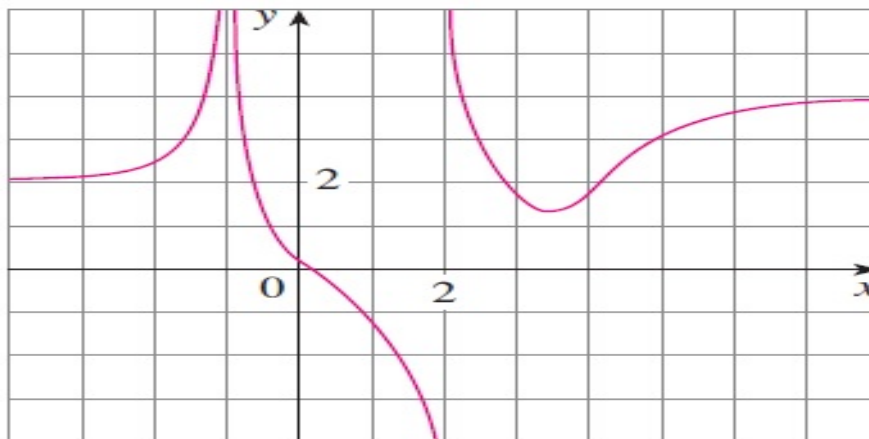
$$f(x) = \begin{cases} x^2 - \ln(x + e), & \text{if } -1 < x \leq 0, \\ e^x - \sin x, & \text{if } x > 0. \end{cases}$$

- (a) Find and classify the discontinuities of f as removable, infinite, or jump.
- (b) Does $\lim_{x \rightarrow 0} (f(x))^2$ exist? explain why or why not?

7. [10 pts.] Use the definition of the derivative to find $f'(1)$ of $f(x) = x|x|$.

8. [10 pts.] Find an equation of the tangent line to the graph of the function $f(x) = \frac{e^x}{x+1} + x + 3$ at $x = 0$.

1. [10 pts.] Use the given graph of a function f shown below to find all vertical and horizontal asymptotes to the graph of f .



The vertical asymptotes are $x = -1, 2$ and the horizontal asymptotes are $y = 2, 4$.

2. [10 pts.] Find the exact value of $\sin \left(\sec^{-1} \left(\frac{5}{3} \right) \right)$.

We let $u = \sec^{-1} \left(\frac{5}{3} \right)$, then $\sec u = \frac{5}{3}$. Now $\sin \left(\sec^{-1} \left(\frac{5}{3} \right) \right) = \sin u = \frac{4}{5}$.

3. [10 + 5 = 15 pts.] Let $f(x) = \ln \left(\frac{2}{x} + 3 \right)$.

(a) Find the inverse function of f .

(b) Find the range of f .

We let $y = \ln \left(\frac{2}{x} + 3 \right)$, then $e^y = \frac{2}{x} + 3$. It is clear that $\frac{2}{e^y - 3} = x$. Therefore, $f^{-1}(x) = \frac{2}{e^x - 3}$. Now the range of the function f is the domain of the inverse function that is $\mathbb{R} \setminus \{\ln 3\}$.

4. [10 + 10 = 20 pts.] Evaluate each of the following limits, if it exists.

(a) $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x^2 - 9x}$

(b) $\lim_{x \rightarrow 1} |x - 1| \sin \left(\frac{1}{x^2 - 1} \right)$

(a) $\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x^2 - 9x} = \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x(x - 9)} \times \frac{\sqrt{x} + 3}{\sqrt{x} + 3} = \lim_{x \rightarrow 9} \frac{(x - 9)}{x(x - 9)} = \lim_{x \rightarrow 9} \frac{1}{x(\sqrt{x} + 3)} = \frac{1}{54}$.

(b) We have $-1 \leq \sin \left(\frac{1}{x^2 - 1} \right) \leq 1$. Thus, we obtain $-|x - 1| \leq \sin \left(\frac{1}{x^2 - 1} \right) \leq |x - 1|$.

It is easy to see that $\lim_{x \rightarrow 1} -|x - 1| = 0 = \lim_{x \rightarrow 1} |x - 1|$. Therefore, by Squeeze Theorem $\lim_{x \rightarrow 1} |x - 1| \sin \left(\frac{1}{x^2 - 1} \right) = 0$.

5. [10 pts.] Find the value of the number A , if any, that makes the function f continuous at $x = 0$.

$$f(x) = \begin{cases} \frac{1}{(x+2)^2} - \frac{1}{4}, & \text{if } x \neq 0, \\ 2 - A \sec x, & \text{if } x = 0. \end{cases}$$

The function is continuous at $x = 0$ if $\lim_{x \rightarrow 0} f(x) = f(0)$. Now it is clear that $f(0) = 2 - A$ and

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{\frac{1}{(x+2)^2} - \frac{1}{4}}{x} = \lim_{x \rightarrow 0} \frac{-x-4}{4(x+2)^2} = \lim_{x \rightarrow 0} \frac{-x^2-4x}{4x(x+2)^2} = \frac{-1}{4}.$$

Therefore, the function is continuous at $x = 0$ if $A = \frac{9}{4}$.

6. [10 + 5 = 15 pts.] Let

$$f(x) = \begin{cases} x^2 - \ln(x+e), & \text{if } -1 < x \leq 0, \\ e^x - \sin x, & \text{if } x > 0. \end{cases}$$

(a) Find and classify the discontinuities of f as removable, infinite, or jump.

(b) Does $\lim_{x \rightarrow 0} (f(x))^2$ exist? explain why or why not?

We have $f(0) = -1$, $\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} (e^x - \sin x) = 1$ and $\lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} (x^2 - \ln(x+e)) = -1$. Therefore, the function f is discontinuous at $x = 0$ since the limit does not exist. The type of discontinuity is jump since $\lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$. The limit in part (b) exists as shown below: we have $\lim_{x \rightarrow 0^+} (f(x))^2 = \lim_{x \rightarrow 0^+} f(x) \times \lim_{x \rightarrow 0^+} f(x) = 1$ and $\lim_{x \rightarrow 0^-} (f(x))^2 = \lim_{x \rightarrow 0^-} f(x) \times \lim_{x \rightarrow 0^-} f(x) = 1$. Therefore, $\lim_{x \rightarrow 0} (f(x))^2 = 1$.

7. [10 pts.] Use the definition of the derivative to find $f'(1)$ of $f(x) = x|x|$.

$$\text{We have } f'(1) = \lim_{x \rightarrow 1} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{x|x| - f(1)}{x - 1} = \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x \rightarrow 1} x + 1 = 2.$$

8. [10 pts.] Find an equation of the tangent line to the graph of the function $f(x) = \frac{e^x}{x+1} + x + 3$ at $x = 0$.

We have $f'(x) = \frac{xe^x}{(x+1)^2} + 1$. It is clear that $f(0) = 4$ and $f'(0) = 1$. Therefore, an equation of the tangent line to the graph of the function f at $x = 0$ is given by $y = x + 4$.